Case Study Lecture

The goal of this case study is to apply the skills you have learned and develop new ones while working with a real-world dataset. Besides the technical side you should also present a business case, meaning in your presentation you should sell us what your group have developed.

Groups

Ideally, the group size should consist of 4 people. If this is not possible, we allow group sizes of 3-5.
In addition, no group should consist solely of members with little or no prior knowledge, regarding machine learning. Please ensure that this is not the case. The groups will be formed in the second week (April 29), at which point we can make adjustments if the prior knowledge is poorly distributed in your group.

Dataset Overview

We are using an Airbnb dataset containing data about the listing, reviews to the listings and calendar including information about prices at specific dates for different cities. There is a general part, which all groups have to do. In addition, your group pick one of the focus areas below. In general the main goal is not on achieving the most accurate results, but rather on the following aspects:

- **Data Preparation:** How do you clean and preprocess the data?
- **Feature Extraction:** What insights can you derive from the dataset? Which are important for your goal?
- **Feature Selection**: What features are actual important?
- **Method Selection:** Which approach best fits your goal?
- **Reasoning:** Why did you choose a specific method?

Before analyzing the data, it is crucial to understand what kind of data you are working with. For example, take a look at how Airbnb incorporates pricing: Airbnb Help Center. We expect that you dive into the data and make decision based on this information.

Download Data

The data is provided by Inside Airbnb https://insideairbnb.com/get-the-data/

Preselected Cities (If you are group 4 you take Paris etc., every group should have its own city):

1. Berlin

2. Amsterdam
3. Barcelona
4. Paris
5. Stockholm
6. London
7. Rome
8. Budapest
9. Copenhagen
10. Istanbul
11. Lisbon
12. Oslo
13. Prague

General Task: Data Preparation

Prepare and explore the Data Set. What additional information can be extracted from the dataset? Possible approaches include:

- **Handling Missing Data:** Identify and manage gaps in the dataset.
- **Reviews Analysis:** Extract meaningful insights from user reviews.
- **Geolocation Analysis:** Identify different areas within a city (e.g., using clustering).
- **Time-Based Patterns:** Analyze booking or usage patterns (clustering may also be helpful here).
- **Additional Data**: You might want to collect additional data specific for your town (e.g., important events, regularity changes)

Possible Focus Areas

> **Choose one of the following Areas**

1. Price Prediction

Which algorithm should you choose? This is not an easy question and depends on the goal you are trying to answer. Often multiple algorithms are possible.

- **Splitting Data**
  - Are there clusters which should only be in training or test dataset?
- **Model Comparison:**
  - Evaluate different prediction models, including different metrics and scores.
- **Hyperparameter:**
  - Different method of finding good hyper-parameters

> Use the `scikit-learn` Pipline

> ⚠ **Use at least three different methods!**

2. Classification

The data set allows for multiple classifications, choose one binary and one multiclass classification.

- **Binary Classification:**
  - Superhost
  - Security Deposit
  - Instant Bookable
  - Require guest phone verification
- **Multiclass Classification:**
  - Cancellation policy
- **Splitting Data**
  - Are there clusters which should only be in training or test dataset?
- **Model Comparison:**
  - Evaluate different prediction models, including different metrics and scores.
- **Hyperparameter:**
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3. Model Evaluation and Comparison of Validation Methods

Evaluating machine learning models is essential to ensure their reliability and performance. This project focuses on exploring and comparing different validation techniques:

- **Cross Validation:**
  - Research about simple and more advances train-test splits (including cross-validation techniques
- **Train-Test Splits:** Different proportions and strategies.
- **Approach Comparison**

The goal is to analyze the differences in performance metrics across these methods and understand their advantages and limitations. You can choose the price prediction or one of the classification problems.

4. Content-Based Recommender System

Often platforms like Airbnb need to recommend the users specific listing. This is a typical challenge for such platforms.

- **Cold start issue**

    - Content-based recommendations need information about a user, e.g., viewings, purchasing history or even likes/ratings.
    - The data don't provide this information directly:
        - You might need to simulate this information.
        - Since reviews of actual users do exist you can try to build profiles based on the reviews.
- **Similarity and Ranking:**
    - Based on the information of the user recommend the three best listings fitting their preferences

5. Be Creative

We are very much looking forward to alternative ideas and would like to motivate you to think about alternatives.
However, please always discuss this with us so that we can ensure that the workload corresponds to that of the other groups.

Evaluation and Grading

The evaluation will account for **40% of the final grade**. The grading is primarily based on the **presentations** and on your submission.

- The **presentations** (22.07. or 23.07.) should clearly present your business case and explain the chosen approach, key findings, and justifications for the methods used.
- The **code notebooks** should provide concise explanations of the methodology and reasoning behind specific choices. It should be well-documented to reflect the thought process and effort invested by the group. Please upload your code notebook until 21.07.2026 23:59.

The presentation will be AFTER the submission

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